

# Aditya Khadkikar

Full Stack AI Engineer | Data Scientist

Determined to delve deeper into learning and building solutions in Deep Learning and AI Engineering.

**Languages:** English, Swedish

## Skills

**Machine Learning:** Linear Regression, Classification, Ensembling Methods

**Deep Learning:** Feed-Forward NN, CNN, ResNet, Graph NN

**Data Science:** Time-Series Analysis, Deterministic/Stochastic Modelling, Markov Chains, Bayesian Density Estimation

**ML/AI Tools:** PyTorch, LangChain (Agentic Framework), TensorFlow, HuggingFace, Flower (FedML), OpenCV

**Data Engines:** PostgreSQL, MongoDB, PySpark, Apache Pulsar

**Cloud Platforms:** AWS, Google Cloud, Openstack

**Web:** Django, VueJS, React, Node.js, Express.js

**Testing:** JUnit, PyTest, Postman

**CI/CD:** Kubernetes, GitLab CI, Ansible, Docker, Docker Compose

## Programming Languages

Python, Java, C, Typescript, HTML/CSS, SQL, Shell Scripting

## Work Experience

**R&D Summer Intern – Energy Efficiency**      June 2026 – Present  
Ericsson R&D – Stockholm, Sweden

Working on energy efficiency of 5G network services by implementing energy-aware, dynamic autoscaling techniques in Kubernetes.

**Development Team – Board Member**      Aug. 2025 – May 2026  
UU AI Society – Uppsala, Sweden

Participated in collaborative event planning and ideation with other board members for organizing networking and informational events with leading companies and startups in the AI innovation space for students at Uppsala University (see <https://www.uuais.com/about>).

**R&D Summer Intern – AI**      June 2024 – Aug. 2024  
Ericsson R&D – Stockholm, Sweden

Worked with orchestration of LLMs in applications of telecommunication products specs-information retrieval. Used LangChain and LangGraph to bring forward a multi-agent RAG solution in Python, evaluating completeness of user request, and retrieves product data, redirecting to different data sources depending on the nature of the query.  
– Used Apache Jena (Fuseki) as a knowledge graph database (schema understanding, query modifications) and OpenSearch as a vector database.

**Teaching Assistant**      June 2023 – Oct. 2023  
Chalmers University of Technology – Gothenburg, Sweden  
• DIT345 Fundamentals of Software Architecture

**Teaching Assistant**      Aug. 2022 – June 2023  
University of Gothenburg, Sweden  
Courses in which I assisted in:  
• DIT185 Software Analysis and Design  
• DIT033 Data Management  
• DIT092 Mini Project – Team Programming  
• DIT023 Mathematical Foundations for Software Engineering

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|----------|-------------------------------------------------------------------------------------------------------------|
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| GitHub   | <a href="https://github.com/adityak714">https://github.com/adityak714</a>                                   |
| Website  | <a href="https://adityak714.github.io/">https://adityak714.github.io/</a>                                   |

## Education

### M.Sc. Data Science – Machine Learning & Statistics

Uppsala University, Aug. 2024 – Jun. 2026

**Completed Thesis Topic:** “Federated Learning in Automated ECG Classification”. Self-authored.

### B.Sc. Software Engineering & Management

University of Gothenburg, Aug. 2021 – June 2024

Joint Department with Chalmers University of Technology.

**Thesis Topic:** “Exploring Metrics for Safety & Reliability Assessment in an Automotive Emergency Braking Perception System”. Achieved **Pass with Distinction**, co-authored with 1 peer.

## Academic Projects

### Bayesian Inference in MRI Diffusion Tensor Imaging

<https://github.com/adityak714/mri-bayesian-inference-methods>

Contributed in a probabilistic posterior analysis project on the Stanford HARDI MRI dataset using stochastic and deterministic approximation methods by *Rokem et al., 2015*. I implemented the Metropolis-Hastings algorithm to approximate the posterior distributions of baseline signal values, diffusivity, anisotropy, and signal angle in a single voxel of brain tissue using Scipy and Numpy.

### Investigating Bias in Facial Recognition in different Race, Age and Gender Demographics

<https://github.com/adityak714/impact-of-bias-in-fr>

Worked on analyzing image recognition models like MTCNN, DLib and Retinaface for correlations between headpose, facial representations and demographic misclassifications, using Python and the Deepface Toolkit for Image Recognition. Listed in Uppsala University’s **EB-Crime Projects**.

### HandyApp: Full-Stack Web Application to find Handymen

<https://github.com/adityak714/mevn-app-handyman-finder>

The full-stack web application bridges the gap between clients who are in need of a handyman for most common reparation and maintenance/installation jobs in homes (plumbers, electricians etc.). Users can send requests as well as publish reviews. Implemented using MongoDB, Express.js, VueJS and Node.

### Marketpulse: Full-Stack AI Stock Market Trend Analysis Application

<https://github.com/adityak714/marketpulse>

Trained a deep neural network on analyzing stock-based tweets to derive sentiment of the future trend of a stock. Worked on creating the admin-side interface, where new models can be uploaded, performance graphs can be generated, and use models as starting points to train new versions. Built using the React Library (TS), Django (Python), with model training using TensorFlow, and deployed using Google Kubernetes Engine.

### DevOps Engineering for Steering Request Algorithm in a Model Car

<https://github.com/adityak714/cpp-cyberphysical-steering-algo>

An algorithm was developed to compute the steering actuation request depending on a small-scaled car’s detected surroundings, and information such as angular velocity along a track. I implemented an end-to-end pipeline that automates the testing of the algorithm on recorded video frames of the car, generate accuracy statistics, version and package the algorithm to be used on a Raspberry Pi with Docker, Bash scripting and GitLab CI.

### Dentistimo: Distributed Dentist Appointment Manager

<https://github.com/adityak714/distrb-systems-ui>

Distributed full-stack dentist appointment management application made with React, MQTT as a publish-subscribe protocol, Typescript, and MongoDB, where users can concurrently manage bookings and book timeslots.